Medical Platform: Project Report

Project analysis report · October 2026

1. General Project Idea

The Medical Platform is a web application that digitizes the daily work of a medical practice or clinic. It replaces paper files, spreadsheets and phone-based booking with one system where patients, doctors and administrative staff work on the same shared data.

The platform covers the full patient journey: a person is registered, books an appointment, is seen by a practitioner, receives a diagnosis and prescription, and has their reports and scans archived in a single record. Every step leaves a trace that can be consulted later in the patient's history.

Main goals

- Centralize patient identity and health background in a single source of truth.
- Simplify appointment booking and reduce no-shows through automated reminders.
- Give practitioners structured tools for consultations, diagnoses and prescriptions.
- Keep medical documents organized and linked to the right patient and visit.
- Protect sensitive medical data through role-based access and an audit trail.
- Stay extensible: a diagnostic analysis module (for example computer vision on medical images) can be plugged in later without changing the core.

Technical foundation

The backend is built with Spring Boot (layered MVC architecture) on top of a PostgreSQL database. Reference values such as gender, blood type, appointment status and document type are stored in lookup tables rather than hard-coded in the code, so they can evolve without redeployment. Authentication is implemented in the last development stages, which is why patient and practitioner profiles are kept independent of user accounts and linked to them later.

2. Project Modules

The system is split into independent modules. Each one has a clear responsibility and communicates with the others through services, never through direct access to their data.

#	Module	Responsibility
M1	Access & Identity Management	Accounts, login, roles, permissions and audit trail
M2	Patient Directory & Demographics	Patient identification, contact details and health background
M3	Scheduling & Appointment Engine	Practitioner availability, booking workflows and visit lifecycle
M4	Clinical Consultations & Prescriptions	Encounter notes, diagnoses, prescriptions and follow-up
M5	Document & Media Archive	Storage of lab results, scan reports and attachments
M6	Diagnostic Analysis (future)	Plug-in analysis of uploaded images and documents
S1	Reference Data (supporting)	Lookup values used by all modules

M1 is built last. M6 is designed now but delivered after the core platform is stable.

3. Module Functionalities

M1: Access & Identity Management

ID	Functionality	Description
1.1	Account creation	Staff accounts are created by an administrator; patients can self-register
1.2	Authentication	Login, logout and password reset
1.3	Role and permission management	Assign roles and control what each role can see and do
1.4	Profile linking	Link a user account to a patient or practitioner profile
1.5	Audit trail	Record logins, record access and profile changes, with search and review

ID	Functionality	Description
1.6	Own account management	Users update their own password and account settings

M2: Patient Directory & Demographics

ID	Functionality	Description
2.1	Create and update patient profile	Identity, contact details, emergency contact, blood type and insurance
2.2	Search patients	Find patients by name, date of birth, phone or national ID
2.3	View patient profile	Display demographic and contact information
2.4	Manage health background	Chronic conditions, allergies and long-term medications
2.5	Patient history timeline	Chronological summary of visits, diagnoses and treatments
2.6	Archive patient record	Soft delete: the record is hidden but never physically erased

M3: Scheduling & Appointment Engine

ID	Functionality	Description
3.1	Manage practitioner profiles	Specialty, license number and active status
3.2	Define working hours	Weekly hours, breaks and slot length per practitioner
3.3	Block time off	Vacations and unavailable periods
3.4	View available slots	Free time slots computed from schedules and existing bookings
3.5	Request an appointment	A patient asks for a slot with a practitioner

ID	Functionality	Description
3.6	Book on behalf of a patient	Staff create an appointment directly, with overlap protection
3.7	Confirm and reschedule	Validate a request or move it to another slot
3.8	Cancel appointment	Cancel with a recorded reason
3.9	Track visit status	Scheduled, Confirmed, In Progress, Completed, Cancelled or No-Show, with allowed transitions only
3.10	View agenda	Daily and weekly calendar view
3.11	Automated notifications	Visit reminders for patients and daily summaries for practitioners

M4: Clinical Consultations & Prescriptions

ID	Functionality	Description
4.1	Open consultation	Start an encounter from an appointment
4.2	Record encounter notes	Chief complaint, physical findings and final diagnosis
4.3	Issue prescription	Medications with dosage, frequency and duration
4.4	Define follow-up plan	Follow-up date, routine checks and care instructions
4.5	Close consultation	Finalize and lock the encounter
4.6	View consultation records	Read past encounters and diagnoses
4.7	View prescriptions	Read and print issued prescriptions

M5: Document & Media Archive

ID	Functionality	Description
5.1	Upload document	Attach a lab result, scan report or PDF to a patient

ID	Functionality	Description
5.2	Classify document	Assign a document type
5.3	View and download documents	Open files linked to a patient record
5.4	Link to consultation	Attach a document to a specific encounter
5.5	Archive document	Remove a document from active views while keeping its trace
5.6	Storage and integrity checks	Store files safely and verify them with a checksum

M6: Diagnostic Analysis (future)

ID	Functionality	Description
6.1	Queue for analysis	Images or documents enter the analysis queue automatically on upload or on request
6.2	Track processing status	Queued, Processing, Analysis Ready or Failed
6.3	Perform analysis	The external engine extracts features and produces screening results
6.4	View insights	Annotations and preliminary results shown beside the consultation notes
6.5	Validate or dismiss findings	The practitioner accepts or rejects each automated result

S1: Reference Data (supporting)

ID	Functionality	Description
S1.1	Manage lookup values	Add, edit or deactivate values such as specialties and document types
S1.2	Read lookup values	Fill dropdowns and validate input in every module

4. Project Actors

Actor	Type	Description
Patient	Human	A person receiving care. Books appointments and consults their own record
Practitioner	Human	A doctor or other clinician. Conducts consultations and owns the clinical content
Medical Assistant	Human	Administrative and reception staff. Manages patients' practical information and the agenda
System Administrator	Human	Technical and security manager. Manages accounts, roles and reference data
System (scheduler and services)	Automated	Background jobs that send reminders, process uploads and check file integrity
Diagnostic Engine	External system	Future component that analyses images and returns results

Access boundaries

- Patients see only their own data.
- Practitioners see the clinical information of the patients they treat.
- Medical assistants work with demographics and the agenda, but do not read clinical notes, diagnoses or health background.
- The administrator manages the system itself and has no routine access to clinical content. Their actions are covered by the audit trail.

5. Assignment of Functionalities to Actors

Legend: ● full access · ◐ restricted (own data, a subset of fields, or a limited action) · – no access

M1: Access & Identity Management

ID	Functionality	Patient	Practitioner	Assistant	Admin	System
1.1	Account creation	◐	–	–	●	–
1.2	Authentication	●	●	●	●	–
1.3	Role and permission management	–	–	–	●	–
1.4	Profile linking	–	–	–	●	–
1.5	Audit trail	–	–	–	●	●
1.6	Own account management	●	●	●	●	–

M2: Patient Directory & Demographics

ID	Functionality	Patient	Practitioner	Assistant	Admin	System
2.1	Create and update patient profile	◐	◐	●	–	–
2.2	Search patients	–	●	●	–	–
2.3	View patient profile	◐	●	◐	–	–
2.4	Manage health background	◐	●	–	–	–
2.5	Patient history timeline	◐	●	–	–	–
2.6	Archive patient record	–	–	–	●	–

M3: Scheduling & Appointment Engine

ID	Functionality	Patient	Practitioner	Assistant	Admin	System
3.1	Manage practitioner profiles	–	◐	–	●	–
3.2	Define working hours	–	◐	●	●	–
3.3	Block time off	–	◐	●	●	–
3.4	View available slots	●	●	●	–	–
3.5	Request an appointment	●	–	–	–	–
3.6	Book on behalf of a patient	–	●	●	–	–
3.7	Confirm and reschedule	◐	●	●	–	–
3.8	Cancel appointment	◐	●	●	–	–
3.9	Track visit status	–	●	●	–	–

ID	Functionality	Patient	Practitioner	Assistant	Admin	System
3.10	View agenda	◐	◐	●	–	–
3.11	Automated notifications	–	–	–	–	●

M4: Clinical Consultations & Prescriptions

ID	Functionality	Patient	Practitioner	Assistant	Admin	System
4.1	Open consultation	–	●	–	–	–
4.2	Record encounter notes	–	●	–	–	–
4.3	Issue prescription	–	●	–	–	–
4.4	Define follow-up plan	–	●	–	–	–
4.5	Close consultation	–	●	–	–	–
4.6	View consultation records	◐	●	–	–	–
4.7	View prescriptions	◐	●	–	–	–

M5: Document & Media Archive

ID	Functionality	Patient	Practitioner	Assistant	Admin	System
5.1	Upload document	◐	●	◐	–	–
5.2	Classify document	–	●	◐	–	–
5.3	View and download documents	◐	●	–	–	–
5.4	Link to consultation	–	●	–	–	–
5.5	Archive document	–	●	–	–	–
5.6	Storage and integrity checks	–	–	–	–	●

M6: Diagnostic Analysis (future)

ID	Functionality	Patient	Practitioner	Assistant	Admin	System / Engine
6.1	Queue for analysis	–	●	–	–	●
6.2	Track processing status	–	●	–	–	●
6.3	Perform analysis	–	–	–	–	●
6.4	View insights	–	●	–	–	–

ID	Functionality	Patient	Practitioner	Assistant	Admin	System / Engine
6.5	Validate or dismiss findings	–	●	–	–	–

S1: Reference Data

ID	Functionality	Patient	Practitioner	Assistant	Admin	System
S1.1	Manage lookup values	–	–	–	●	–
S1.2	Read lookup values	●	●	●	●	●

Summary by actor

- **Patient:** self-registration, booking requests, viewing and managing their own appointments, and read access to their own profile, consultations, prescriptions and documents.
- **Practitioner:** the clinical core (M4), full access to patient history and documents, their own agenda and, later, the diagnostic insights (M6).
- **Medical Assistant:** patient registration and contact data, the whole agenda and visit flow (M3), and document upload without access to clinical content.
- **System Administrator:** accounts, roles, audit trail, practitioner records, reference data and archiving.
- **System and Diagnostic Engine:** notifications, file integrity checks and, in the future, automated image analysis.

Assumptions to validate

- Patients can self-register and upload their own external reports.
- Medical assistants do not see allergies or any health background. If reception needs them (for example for a safety alert), this should be changed to restricted access.
- Patients see a consultation only after it is closed by the practitioner.
- The administrator archives patient records, but a clinical retention policy should be defined before deletion rules are final.